

Study Exploring Security Threats in Waste Phones

A Life cycle Based Approach

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Abstract—Waste Electrical and Electronic Equipment (WEEE) recycling has become imperative for every nation where the large scale WEEE e-waste management is considered crucial, however, often neglecting small WEEEs. Mobile phone is one of the most important and widely used small WEEE. High metal concentration of mobile phones make their systematic recycling of paramount importance; in contrast, a very small fraction of these devices end up in the government sanctioned recycling units. A majority of the devices are refurbished and repaired which enters the second-hand device market. The existence of the second-hand market has often been neglected in the supply chain of WEEE. Thus, the second-hand device market has introduced the problem of improper channeling and disposal of waste mobile phones with disoriented e-waste management. This can lead to potential threat of exposing personal information from the device of the owner, including login credentials of crucial applications, emails, contact list, messages, and photos. Such disclosures can lead to security threats such as identity theft, data tampering, information disclosure, and others. In this study, through systematic background review, we provide theoretical characterization of the small WEEE, develop the life cycle framework of a mobile phone, identify potential security threats of the devices including cloud storage, and propose mitigation techniques in the implementation stage within the supply chain.

Index Terms—Waste Mobile Phones, E-waste, Online Security, Security Threats, Supply Chain, Mobile Phone Recycling, WEEE, Cloud Storage.

I. INTRODUCTION

Mobile phones have become an integral part of everyday interactions with a paradigm shift to smart phones from feature phones resulting in reduced device lifespan [21]. Intelligent marketing strategies, lucrative features, and technological advancement has successfully tailored the desire of the consumers, however reducing the average lifespan of mobile phones from 10 years to 2 years on an average [24]. The global smart phone install base has increased to four billion as per 2016 reports and is predicted to reach six billion by 2020 [1]. Defunct mobile phones thus are increased in bulk and contribute to electronic waste (e-waste) and demands proper environmental control measures through recycling and refurbishing [52, 59]. Such devices are classified as Waste Electrical and Electronic Equipment (WEEE).

The waste mobile phones are rich source of different metals and Rare Earth Elements (REE) [27]. Due to the high metal and REE concentration mobile phone devices contribute largely for urban mining [67]. However, the current supply

chain system lacks the rigor for successful urban mining and waste disposal of defunct mobile phones and we require more efficient supply chain management schemes for defunct mobile phones or small WEEEs in general. The existing supply chain is a closed loop and often have other neglected secondary branch loops.

In countries such as UK, Switzerland, USA, Canada, and others, proper e-waste management schemes are adapted which prohibits branched loops such as informal sectors or second-hand market [21]. In such countries, the mobile phones are either sold after use, exchanged, or sent to the recyclers [33]. Many companies follow proper e-waste management schemes, for example, Apple Inc. uses systematic management for its used mobile phone which enables the recovery of valuable materials. Apple has recently, introduced their latest robot named Daisy for efficient dismantling and segregation of different valuable parts of iphone with an aim to reuse them again which is an important step towards circular economy [2]. In developing countries, such as India, Thailand, Bangladesh, and others, in addition to the primary interaction between a company and their customers, there is a huge market for secondary and refurbished electronics [33, 41]. Several repair and refurbish cafe exists in such countries which repair mobile phone without license and thus introduce the potential security risks of personal data of the owners [65, 25].

E-waste literature is replete with examples of recycling, policy technological issues, sustainability [11, 22] and to some extent security threats of mobile phone devices [10, 29]. However, security threats of used, recycled, or refurbished mobile phones is still a relatively unexplored area. The primary objective of this study is to identify the security threats of the defunct mobile phones including that of the vulnerabilities of cloud storage and improper data storage and access management. We further develop a life cycle of a mobile phone device relevant to developing countries by introducing the secondary branches of material flow involving informal and second-hand market. We propose modification of the existing supply chain for better management of defunct mobile phone devices and small WEEEs in general.

II. RELATED WORK

Data security from e-waste is an area where very few interventions have been done in the past. The first possible publication which introduced a classification of hardware reverse

engineering steps and mitigation techniques by McLoughlin addressed the security threats arising from reverse engineering in secured embedded systems [50]. Quadir et al. presented a review on reverse engineering (RE) techniques to exploit security vulnerabilities of used devices through Chip level RE; Printed Circuit Board (PCB) level RE, and system level RE [54]. Grand has described physical, chemical and soft tool methods of removing layer of printed circuit boards with degrees of difficulty and cost involved for reverse engineering of PCB [35].

Torrance and James provided detailed methodologies followed in the semiconductor industry for reverse engineering of integrated circuits accompanied by case studies [63]. It was first mentioned in 2009 in a research report by CSO that e-waste disposal attitude is dictated by data security not by environmental concerns [3]. The security threats arising from electronic waste discussed by Roychowdhury et al. focused on the electronic components and PCBs for which they discussed the potential security threats and provided solutions [58]. Following this work Alghazo and Ouda reported the security threats from e-waste in the Gulf Cooperation Council (GCC) countries [8, 9]. It is evident from the previous studies that the focus has been mostly on electronic components, devices, and PCBs. Some interventions have been done on the e-waste data security but that has a more generalized approach where device specific evaluation of security threats is often overlooked.

III. SECURITY THREATS AND DATA MANAGEMENT

Due to the improper existing e-waste management practices, security threats of the e-waste are multifaceted, especially for the defunct mobile phones. Thereafter, several vulnerabilities exist in the currently adapted data management and protection through cloud storage in these mobile phones.

A. Defunct Mobile Phone Security Threats

E-waste has far fetching implications when we discuss about security threats [58]. Studies have shown that the e-waste industry has grown up to \$7 billion [36] and cybercrime industry has financially depleted more than \$450 billion annually worldwide [34, 5]. These two industries are interlaced when proper methods of disposal are not adapted either by the consumers or by the recycling organizations. Reports have been found on organized crime [32] by obtaining information from second-hand mobile phones [4, 39].

Defunct mobile phone devices are of the following types:

- Repaired Devices
- Recycled Devices
- Resaled Devices

These devices contribute to the repository of e-waste and are also part of the second-hand market. second-hand market often lacks proper recycling techniques and management schemes and thus contributes largely in the security threats since they follow unmonitored or loosely monitored data handling practices [46, 15].

Thus, mishandling of device data enables several malicious actors to track personal and unintended information of the users [60]. Personal data include, login credentials [20], contact list [19], messages [26], photos [53], and other crucial information of the owner and their connections. Such personal

data leakage is harmful since such attacks not only reveal personal data but can impact ones contacts, bank details [49], access personal photos. Mobile phones are susceptible to viruses, trojans, worm [17, 28, 43], however such vulnerability of e-waste reduce the need for sophisticated attacks [37, 38, 55] and can be obtained easily. Developing countries (such as Nigeria, Ghana, Pakistan) [7] are often targeted for cybercrime practices [16, 31] and thus the population is severely harmed by such activities. Developing countries contribute majorly in generating online frauds [47] and the e-waste management is also done poorly for such countries [61].

B. Cloud Storage Security Threats

Smart phone data storage has increased in recent years due to the deployment, development, and execution of several applications [48]. In order to mitigate memory shortage problems and proper data management cloud storage is integrated by the developers and the users to regularly perform data backup and to avoid loss of information [6]. Cloud services contribute to sustainable data storage and backup solutions.

Cloud storage provide periodic storage of data by running in the background on the remote servers through network connectivity [66]. It enables transfer of data between devices and between users connected over internet [14]. Thus, the data is not dependent on a device and is in fact ubiquitous in nature facilitating easy and time-effective transfer of data between devices [30, 45]. Cloud based solutions have evolved from online backup services to primary storage services. Increased trend among the consumers to store their data directly in cloud-based repositories instead of their solid state internal storage devices (hard drives) has been noted [14].

Data security is also handled by cloud storage since instead of the actual local file, a dummy file is kept in the local repository. The original data is securely stored in the cloud and it can be accessed or downloaded back onto the device as needed. Required changes to the files are pushed to the cloud so that no local copy is kept on the device. This frequently reduces the risk of data leaks when a device reaches its End-of-Life (EoL) [12]. However, it is important to note that, the cloud storage login password should be changed or updated periodically or as needed. If not updated properly, cloud storage security advantages can modify to disadvantages since the refurbished phones can lead to security threats such as data leakage.

Additionally, backup storage are susceptible to various security attacks such as Man-In-The-Middle attacks, IP spoofing, port scanning, packet sniffing, and others [42, 40]. To reduce the data loss, redundant data are stored in various locations while creating backups which might be problematic from data security standards as well [62]. Similar problems are noted for cloud storage which can be duplicated across multiple cloud service providers.

Privacy is also concerning for cloud storage since user data is stored in cloud. Data encryption for the cloud storage works only if the encryption key is protected such that only the developer can decrypt the stored data. Using encryption tools to encrypt data with ones own key before transferring it into the cloud creates security flaws [44]. Some cloud service providers are now offering this option and letting user choose

their own key [45]. This technique is helpful but is challenging for those with lesser technical knowledge and awareness.

IV. DEVELOPMENTAL RESEARCH

The study is focused on the EoL of mobile phones. The principal interest of the study is to trace the path of mobile phones when they transform into an e-waste. Thus, we focused on the various stakeholders, consumers and the mobile shops including the sellers and repair shops. For this study, we are specifically researched on the less explored second-hand market.

Semi-structured interviews were conducted with randomly sampled mobile phone users within the age limit of 25 to 45 years. While developing the interview questions different online database sources were explored to search relevant literature using appropriate keywords. We noticed that the previous studies have not explored the field of second-hand market while discussing about e-waste and it's security threats. To investigate further on the supply chain of mobile phone the researchers conducted field visits as well to understand the marketing and development procedure. In addition to the mobile phone consumers, we also asked questions to mobile phone sellers as well as repair shops. For the users, the focus was on their disposal attitudes towards waste mobile phone and whether they opt for formal e-waste recycling. While interviewing the mobile phone sellers, the primary focus was on exchange and buy-back schemes. We noted that the mobile

spots and the associated security threats. Debnath et al. [23] discuss about the life cycle of small WEEE and different phases of the life cycle. Based on the aforementioned work, a structured framework of the life cycle of waste mobile phones has been developed and shown in Figure 1.

Figure 1 include stakeholders such as Original Equipment Manufacturers (OEMs). Mobile phone reaches the consumer either via e-commerce or traditional retailer chain. After using it for suitable amount of time (which can vary from 3 months to 3 years), the question arises, whether it is an Used Electrical and Electronic Equipment (UEEE) or it is a Waste Electrical and Electronic Equipment. If it is a WEEE, it is either collected via buy back scheme or by authorized WEEE collectors. Thereafter, a mobile phone device is dismantled and processed by the WEEE recycling facility for material recovery and recycling. If, it is a UEEE, there are several routes, here referred to as back loops, which extends the product life cycle are mentioned below.

- 1) The mobiles may be sold as a second-hand phone via online hosting sites or by personal contact.
- 2) If mobiles are repairable, it is repaired and reused by the same user.
- 3) If the mobile is not functioning properly, it can be exchanged. These phones are then refurbished and sold online as well as in the second-hand market.

We note that back loops are often not recognized and therefore a larger portion of it's contribution is unaccounted and never considered while developing the supply chain model of mobile phone devices. These back loops can be iterated more than once as these phones may be repaired and refurbished multiple times. Once these mobile phones stops functioning properly, they enter the WEEE stream subjected to recycling or refurbishing and are termed as defunct devices.

VI. PROPOSED SOLUTION AKA DISCUSSION

Previous research and our interviews noted that environmental and security threat awareness for e-waste disposal is extremely low and this prevails for different stakeholders. For developed countries, stakeholders though aware of e-waste disposal procedure, however there the lack of security threat awareness is problematic [13, 51]. In this study, we have featured the possible security threats that can emerge over the life cycle of a defunct mobile phone. The back loops in the life cycle framework (Figure 1) represents the pathways where the majority of the security threat exists. The life cycle framework may be treated as a prima facie of what a supply chain network of mobile phone might look like. Security threats can be minimized by active participation of the users, strict policy intervention and most importantly monitoring the vulnerable nodes of the supply chain. In order to minimize the security threats along supply chain the following solutions are proposed.

- 1) The users should be made aware of the possible security vulnerabilities. This enables them to take preventive measures as well as scrutinize their decisions of opting for a third-party refurbisher.
- 2) Strict policies should be adapted, implemented, and periodically monitored to control the activities of the third-

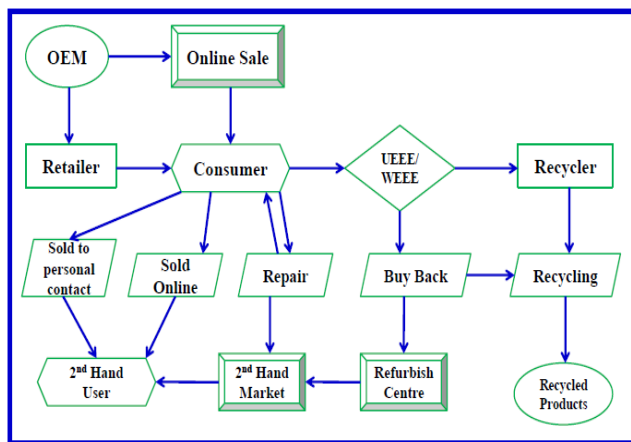


Fig. 1. Life cycle based framework of small WEEE (Mobile Phones)

repair shops were ignorant in answering the questions yet some of them responded with much concern regarding recycling and refurbishing old and used mobile phones. The interviewees were questioned on device repairing issues, types of people who repair, and on how the defunct parts were monitored after they were discarded or implemented in other processes. Based on the answers of the interview we developed a life cycle based framework for waste mobile phones especially focusing on the second-hand market with further analysis.

V. LIFE CYCLE FRAMEWORK DEVELOPMENT OF SMALL WEEE

The life cycle of waste mobile phones is important in facilitating the identification of the back-loops, potential weak

party and unauthorized repair shops which is referred in the paper as second-hand market in general.

- 3) Repairing and reusing is a very common practice in the developing countries. In the developed nations, more repairing cafes are emerging as well [57, 56]. Though both (repair cafes and third-party refurbishers) are conceptually same but repair cafe provides a community service approach. Refurbish cafes may be established in the developing world which may reduce risks of data leaking and smuggling.
- 4) It is noted that the take-back system sends a wrong message to the users and often users are noted to be negligent about such messages [18]. The mobile phone companies should make more efforts to channelize the waste mobile phones and in general e-waste towards recycling plants instead of heuristic approaches adapted in the second-hand market.
- 5) Cloud storage contribute to a safer and more reliable solution for data backup and storage. However, we have noted several security threats as discussed in section III-B. Periodic and regular password management and user awareness should be done.
- 6) The second-hand market is a very active, vibrant and dynamic market which is contributing in regular economic transactions [64]. Thus, the existence of the second-hand market is a boon for the sustainability seekers. On the other hand, the infiltration of the unauthorized repairing shops and refurbishing activities injects various unattended security threats. One solution is to provide such markets and the technicians with proper training from the manufacturers and to franchise such shops by providing them with monitored license. Policy intervention can create synergy with the authorized recyclers as well as ensure circular economy. However, we also need to acknowledge the economic condition of the small shops which belong to the second-hand market.
- 7) The biggest challenge lies in developing an efficient supply chain network for routing the EoL mobile phones towards sustainable 3R solutions. Activities of third-party refurbishers may be restricted to limited repairing activities only. The faulty components can be collected from the third-party recyclers and routed towards recycling facility.
- 8) The repairing industry, proliferated with proper training, license, and recognition, can contribute substantially in enhancing the product life cycle. Risks of data threats will be minimized with policy interventions and periodic monitoring. Dishonest and unrecognized organizations with such activities can be penalized. Periodic and regular monitoring of the shops can ensure that the shops are operating properly and if not their license can be revoked as well. This will ensure proper material flow and minimization of risks.
- 9) Most importantly, the consumers must be aware of proper e-waste disposal. They must dispose their mobile phones which are beyond repair via authorized e-waste recyclers who provide responsible data destruction certificates.

VII. CONCLUSION

In this study, we acknowledge and introduce the existence of second-hand market and discuss the security threats of defunct mobile phones. We identify the potential threats of mobile phone repairing and refurbishing especially from third-party repair shops and refurbishers. We noted the influence of the second-hand market which is a key driver for extension of mobile phone life cycle. The existence of such back loops prohibits the defunct devices to reach the authorized recycling facilities. We note several security threats in general for the defunct mobile phone devices and also highlight the security vulnerabilities of cloud based solutions. A life cycle based framework for mobile phone devices has been developed which can be treated as a *prima facie* for further model development of small and medium WEEEs. Potential solutions for the problem were proposed with a focus on improved supply chain management and policy framework. The current study, nevertheless, has focused on the security threats of the back loops in a device life cycle which is a less explored domain and further studies will certainly enrich studies in sustainability and security research. The findings are expected to be helpful for researchers, relevant stakeholders, and policymakers.

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